

Concave Programming

For positive integers n and m , let $f : \mathbb{R}_+^n \rightarrow \mathbb{R}$, and $g : \mathbb{R}_+^n \rightarrow \mathbb{R}^m$ be continuously differentiable functions. Consider the problem

$$\max_{x \in \mathbb{R}_+^n} f(x) \tag{1}$$

subject to the constraint that $g(x) \leq 0$. Let $\mathcal{L}(x, \lambda) = f(x) - \lambda \cdot g(x)$.

The *Kuhn-Tucker conditions*: there is a point $(x^*, \lambda^*) \in \mathbb{R}^n \times \mathbb{R}^m$ such that

$$\begin{aligned} \mathcal{L}_x(x^*, \lambda^*) &\leq 0 & x^* \cdot \mathcal{L}_x(x^*, \lambda^*) &= 0 & x^* &\geq 0 \\ \mathcal{L}_\lambda(x^*, \lambda^*) &\geq 0 & \lambda^* \cdot \mathcal{L}_\lambda(x^*, \lambda^*) &= 0 & \lambda^* &\geq 0 \end{aligned} \tag{2}$$

The *Saddlepoint condition*: there is a $(x^*, \lambda^*) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ such that

$$\mathcal{L}(x, \lambda^*) \leq \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x^*, \lambda) \quad \text{for all } (x, \lambda) \in \mathbb{R}_+^n \times \mathbb{R}_+^m. \tag{3}$$

Slater's constraint qualification: There is a point $\hat{x} \in \mathbb{R}_+^n$ such that $g(\hat{x}) \ll 0$.

We will use these facts.¹

1. If x^* solves problem (1) and g is *affine* (a constant plus a linear function), then there is a point $\lambda^* \in \mathbb{R}_+^m$ such that the Kuhn-Tucker conditions hold.
2. If x^* solves problem (1), f is concave, g is convex, and Slater's constraint qualification holds, *then* there is a point $\lambda^* \in \mathbb{R}_+^m$ such that the Kuhn-Tucker conditions hold.
3. If f is concave, g is convex, and there is a $\lambda^* \in \mathbb{R}_+^m$ such that the Kuhn-Tucker conditions hold, *then* x^* solves problem (1).
4. If f is quasiconcave, g is quasiconvex, there is a $\lambda^* \in \mathbb{R}_+^m$ such that the Kuhn-Tucker conditions hold, *and* $f_x(x^*) \neq 0$, *then* x^* solves problem (1).
5. If the Saddlepoint Condition holds, then x^* solves problem (1).
6. If x^* solves problem (1), f is concave, g is convex, and Slater's constraint qualification holds, then the Saddlepoint condition holds.

Note that facts 1. and 2. give properties so that the Kuhn-Tucker conditions are *necessary* for a solution; 3. and 4. give properties for the Kuhn-Tucker conditions to be *sufficient* for a solution. I will use fact 3. in the proof for the First Welfare Theorem for the 2-good quasilinear economy (under convexity assumptions on preferences and technologies). Facts 4. -5. are from H. Uzawa, "The Kuhn-Tucker Conditions in Concave Programming," *Studies in Linear and Non-Linear Programming*, 1958, available on the course Canvas page.

¹Facts 2. and 3. are summarized in Theorem 7.16 in R. Sundaram's *A First Course in Optimization*. Note however that he subsumes nonnegativity constraints into his "h" function, whereas I list them separately. Fact 4. is his Theorem 8.13. He proves each Theorem in the corresponding chapter appendix.